LN3 Information and Conditioning

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Outlines

Textbook: Stochastic Calculus for Finance II: Continuous-Time Models: Chapter 2

1. Information and σ-algebra
2. Independence
3. General Conditional Expectations

Information and σ-algebra

Definition 1

Let Ω be a nonempty set. Let T be a fixed positive number, and assume that for each $t \in [0, T]$ there is a σ-algebra $\mathcal{F}(t)$. Assume further that if $s \leq t$, then every set in $\mathcal{F}(s)$ is also in $\mathcal{F}(t)$. Then we call the collection of σ-algebras $\mathcal{F}(t), 0 \leq t \leq T$, a filtration.

Definition 2

Let X be a random variable defined on a nonempty sample space Ω. The σ-algebra generated by X, denoted $\sigma(x)$, is the collection of all subsets of Ω of the form $\{X \in B\}$, where B ranges over the Borel subsets of $\mathbb{R}$.

$\mathcal{G}$-measurable and $\mathcal{F}(t)$-measurable

Definition 3

Let X be a random variable defined on a nonempty sample space Ω. Let $\mathcal{G}$ be a σ-algebra of subsets of Ω. If every set in $\sigma(X)$ is also in $\mathcal{G}$, we say that X is $\mathcal{G}$-measurable.

Definition 4

Let Ω be a nonempty sample space equipped with a filtration $\mathcal{F}(t)$, $0 \leq t \leq T$. Let $X(t)$ be a collection of random variables indexed by $t \in [0, T]$. We say this collection of random variables is an adapted stochastic process if, for each t, the random variable $X(t)$ is $\mathcal{F}(t)$-measurable.

Independence of Two Random Variables

Definition 5

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{G}$ and $\mathcal{H}$ be sub-σ-algebras of $\mathcal{F}$ (i.e., the sets in $\mathcal{G}$ and the sets in $\mathcal{H}$ are also in $\mathcal{F}$). We say these two σ-algebras are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B) \text{ for all } A \in \mathcal{G}, B \in \mathcal{H}.$$

Let X and Y be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. We say these two random variables are independent if the σ-algebras they generate, $\sigma(X)$ and $\sigma(Y)$, are independent. We say that the random variable X is independent of the σ-algebra $\mathcal{G}$ if $\sigma(X)$ and $\mathcal{G}$ are independent.

Independence of More Than Two Random Variables

Definition 6

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3, \cdots$ be a sequence of sub-σ-algebras of $\mathcal{F}$.

(1) *For a fixed positive integer n, we say that the σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \cdots, \mathcal{G}_n$ are independent if*

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2) \cdot \cdots \cdot \mathbb{P}(A_n)$$

for all $A_1 \in \mathcal{G}_1, \cdots, A_n \in \mathcal{G}_n$.

Independence of More Than Two Random Variables

Definition 6 Continued

Let $X_1, X_2, \cdots$ be a sequence of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

(2) We say the n random variables $X_1, X_2, \cdots, X_n$ are independent if the σ-algebras $\sigma(X_1), \cdots, \sigma(X_n)$ are independent.

(3) We say the full sequence of σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \cdots$ is independent if, for every positive integer n, the n σ-algebras $\mathcal{G}_1, \cdots, \mathcal{G}_n$ are independent.

(4) We say the full sequence of random variables $X_1, X_2, \cdots$ is independent if, for every positive integer n, the n random variables $X_1, X_2, \cdots, X_n$ are independent.

Independence

Theorem 1

Let X and Y be independent random variables, and let f and g be Borel-measurable functions on $\mathbb{R}$. Then $f(X)$ and $g(Y)$ are independent random variables.

Proof $A \in \sigma$-algebra generated by $f(X)$. $B \in \sigma$-algebra generated by $g(Y)$. Then $A \in \sigma(X)$, $B \in \sigma(Y)$. Since X and Y are independent, we are done.

General Conditional Expectations

Definition 7

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\mathcal{G}$ be a sub-σ-algebra of $\mathcal{F}$, and let X be a random variable that is either nonnegtive or integrable. The conditional expectation of X given $\mathcal{G}$, denoted $\mathbb{E}(X|\mathcal{G})$, is any random variables that satisfies

(1) *(Measurably) $\mathbb{E}(X|\mathcal{G})$ is $\mathcal{G}$-measurable, and*

(2) *(Partial averaging) $\int_A \mathbb{E}(X|\mathcal{G})d\mathbb{P}(\omega) = \int_A X(\omega)d\mathbb{P}(\omega)$ for all $A \in \mathcal{G}$.*

If $\mathcal{G}$ is the σ-algebra generated by random variable W, i.e., $\mathcal{G} = \sigma(W)$, we generally write $\mathbb{E}(X|W) = \mathbb{E}(X|\sigma(W))$.

Martingale

Definition 8

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let T be a fixed positive number, and let $\mathcal{F}(t), 0 \leq t \leq T$ be a filtration. Consider an adapted stochastic process $M(t), 0 \leq t \leq T$. If $\mathcal{E}(M(t)|\mathcal{F}(s)) = M(s)$ for all $0 \leq s \leq t \leq T$, we say $M(t)$ is a martingale. It has no tendency to rise or fall.

Markov Property

Definition 9

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let T be a fixed positive number, and let $\mathcal{F}(t), 0 \leq t \leq T$, be a filtration. Consider an adapted stochastic process $X(t), 0 \leq t \leq T$. Assume that for all $0 \leq s \leq t \leq T$ and for every nonnegative, Borel-measurable function f, there is another Borel-measurable function g such that

$$\mathbb{E}(f(X(t))|\mathcal{F}(s)) = g(X(s)).$$

Then we say that the X is a Markov process.